

Particulate Matter Monitoring & Health Effects

Issue 2026-10-02 | Entry window 2 October 2026 (full day) | 15 records screened in scope

15

RECORDS IN SCOPE
(56 EXCLUDED)

8

OF 10 SUBTOPIC
BANDS POPULATED

3

EFFECT ESTIMATES
WITH A CI

2

TIER-A
RECORDS

1

RECORDS CARRIED ON
METADATA ALONE

Signal of the day

A 25-year, 100 m daily PM_{2.5} surface for the contiguous US — and, the same day, a rural low-cost network showing where interpolation stops working.

Sheidaei and colleagues (*ES&T*, tier A) fill missing satellite AOD with a reanalysis-informed U-Net, refine it with a bidirectional LSTM, downscale it on terrain to 100 m and predict daily PM_{2.5} over >766 million cells for 2000–2024. Site-held-out validation: $R^2 = 0.82$, $RMSE = 2.85 \mu g m^{-3}$; spatial R^2 0.94, temporal **0.78**. **Caveat:** the 100 m detail is inherited from downscaled AOD and land use, and regulatory monitors cannot validate sub-kilometre gradients; smoke-day skill is not reported. Mielnik and colleagues (Research Square, tier A) ran low-cost sensors at **46** sites in the Everglades Agricultural Area for 18 months: day-by-day leave-one-out R^2 ranged **0 to 0.99**, IDW beat kriging on most days, and smoke added $1.65 \mu g m^{-3}$ (SE 0.13) inland. **Why it matters for sensing:** the national product is smooth where the agricultural plume is patchy; the honest use of a dense LCS network is as the residual on such a surface, and the 0–0.99 range is the size of the day-to-day failure that a single-model exposure surface hides.

What changed today

- **Periconceptional PM₁₀ and congenital heart disease, Delhi NCR** (1,115 cases / 441 controls): post-conception PM₁₀ AOR **3.36** (2.05–5.52) for all CHD and **4.35** (2.41–7.83) for VSD; preconception PM₁₀ **2.82** (1.69–4.71) for simple CHD. ORs of 3–4 are far above the literature; the exposure contrast is not stated and controls are outnumbered 2.5:1.
- **Conflict as an emissions experiment.** Across nine Middle Eastern countries during the 39-day 2026 conflict, column anomalies ordered by lifetime — SO₂ –**45.6 %**, AOD –**19.6 %**, NO₂ –15.1 %, CO –8.6 %, CH₄ flat — point to reduced hydrocarbon-sector emissions. The authors decline a PM_{2.5} burden: dust and anthropogenic aerosol are inseparable at CAMS resolution.
- **Condensables carry the metals.** Stack sampling in Chinese coking plants puts **44.5–57.4 %** of flue-gas trace elements in the condensable fraction, with As and Cd enriched in PM_{2.5}; a 307-plant inventory gives **159.8 kt** PM in 2022, falling to 15.5 kt by 2050 under BACT.
- **Instrument metadata.** For the SwisensPoleno Jupiter, the choice of baseline material (H₂O, NaCl, SiO₂) changes the share of fluorescence-positive particles — NaCl thresholds are up to 15× higher. Bioaerosol counts are not comparable across networks unless baseline and threshold rule are reported.
- **Thin Friday.** PubMed entered only **7** PM records under [EDAT] 2 October; Crossref by ISSN (40) was mostly off-topic *AMT/ACP*, *MST* and *ES&T* water chemistry. One metadata-only record (7%, the series low): an Aveiro-group indoor/outdoor school PM₁₀ toxicology paper in *APR*, on the retry list.

Window continuity

The 1 October issue closed with `last_entry_date` at 1 October. This issue collects **2 October in full**, continuous with it. The run started at 14:50 CDT on 3 October, before the 22:00 cut, so the newest buildable date is 2 October; **3 October and the W40 weekly (27 September–3 October) are left to the next run.**

Corpus analytics

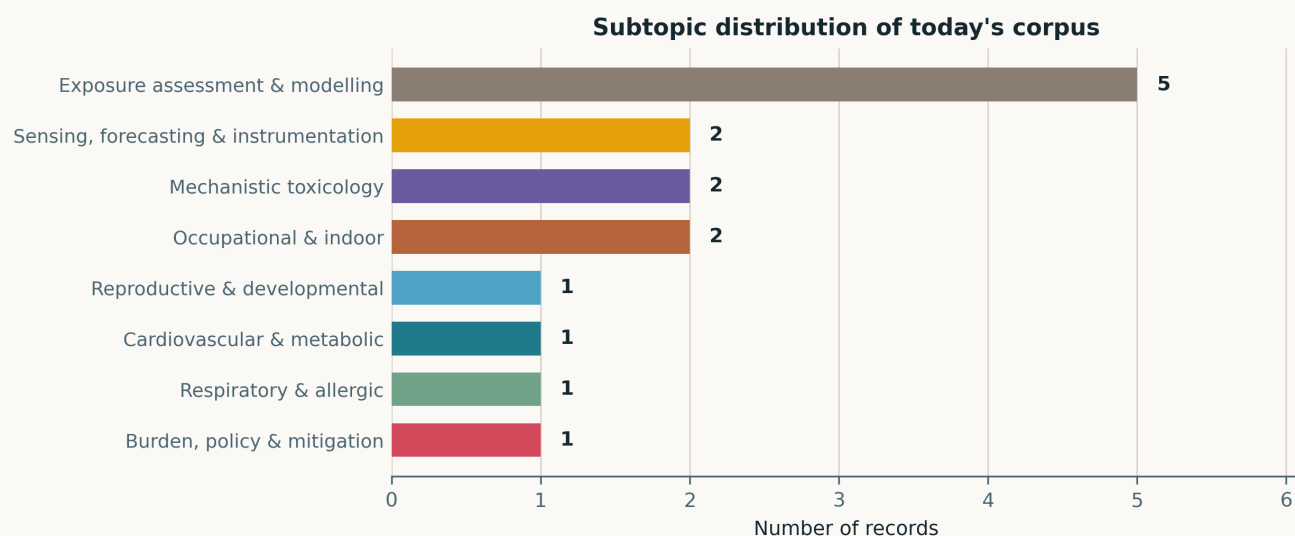


Fig. Subtopic assignment of the 15 in-scope records. **Eight of ten bands populated**; *Exposure assessment* leads with **five** (two tier A), then two each for *Sensing*, *Mechanistic toxicology* and *Occupational & indoor*. *Neuro / mental health* and *Other clinical* are empty; the one neuro-relevant record is an astrocyte study filed under toxicology.

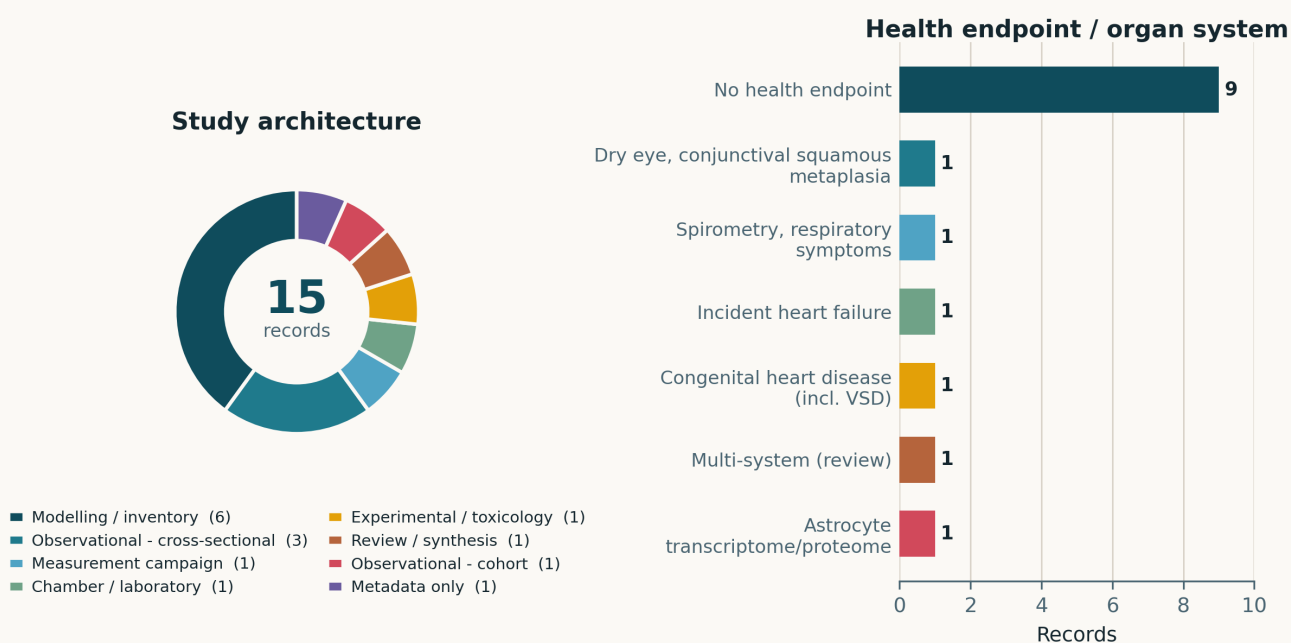


Fig. Left — architecture: modelling / inventory six (two deep-learning models, the CTM, the satellite analysis, the inventory, the interpolation study), cross-sectional three, and one each for the field campaign, laboratory validation, in-vitro exposure, review, cohort and the metadata-only record. Right — **nine** records carry no health endpoint.

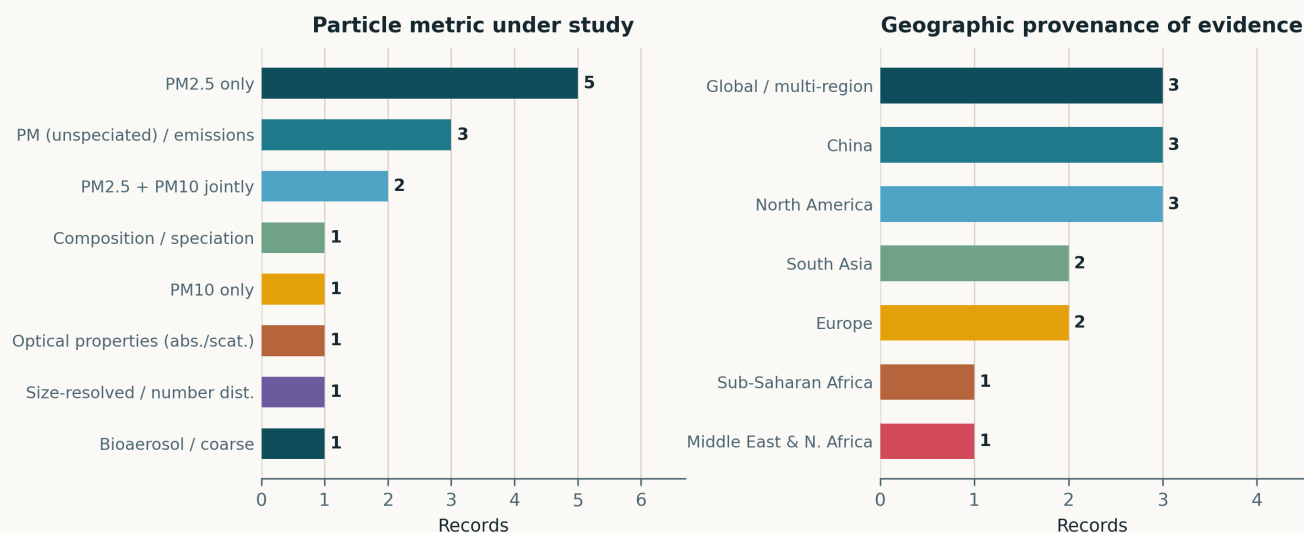


Fig. Left — particle metric: PM_{2.5} only five, unspeciated / emissions three, PM_{2.5} + PM₁₀ two, and one each for PM₁₀, size-resolved (PM_{0.2}), bioaerosol, optical (BC) and composition. Right — geography: North America and China three each, global / not stated three (laboratory, review, metadata-only), Europe and South Asia two each, Middle East and Sub-Saharan Africa one each.

Life-course windows addressed today (records may span several stages)

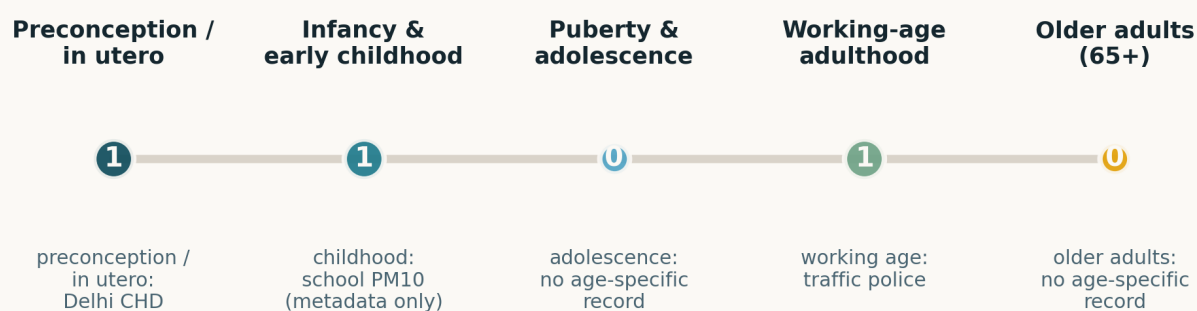


Fig. Life-course windows: in utero one (Delhi CHD), childhood one (school PM₁₀, metadata only), working age one (traffic police), adolescence and older adults none.

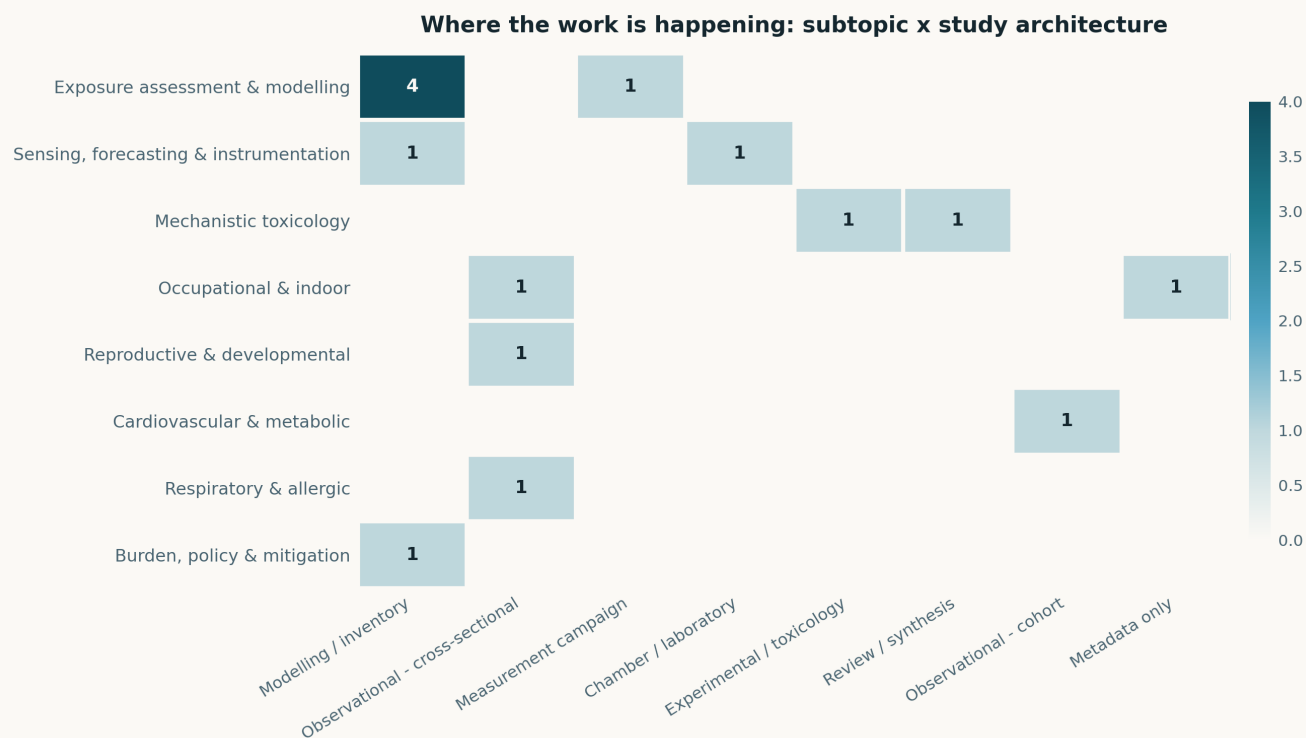


Fig. Subtopic × study architecture for the 15 records. Modelling sits in exposure, sensing and burden; the three cross-sectional records are spread across respiratory, occupational and reproductive (the case-control study).

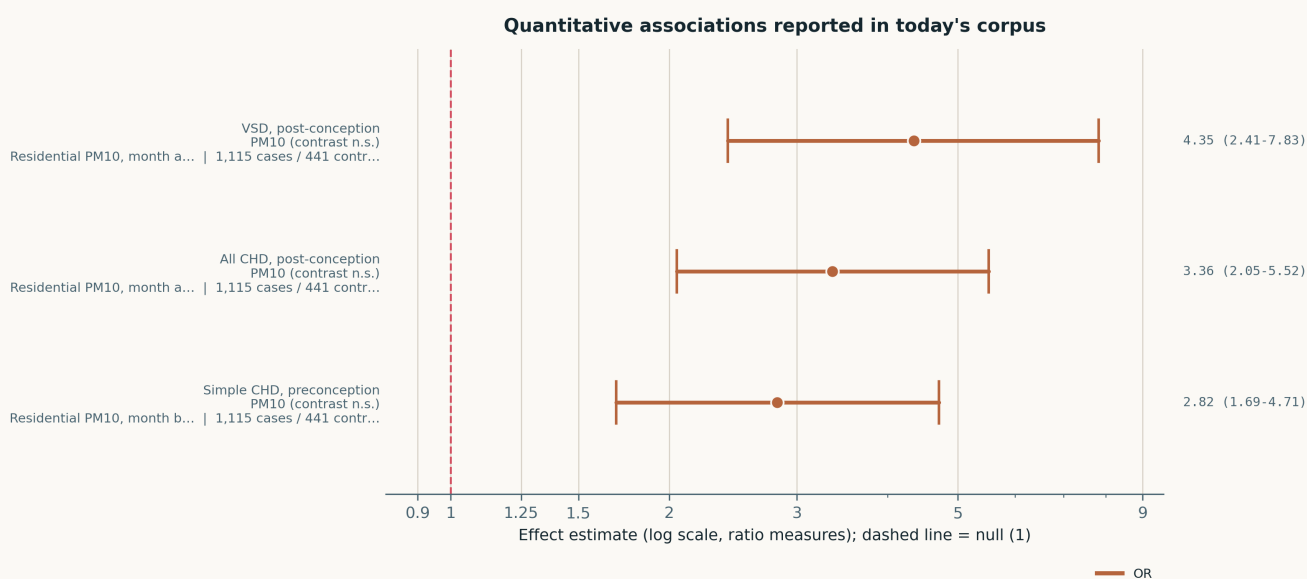


Fig. Three adjusted odds ratios with 95 % CIs, all from the Delhi NCR CHD case-control study, log scale. The PM₁₀ contrast (per unit or category) is not given in the abstract, and the three share cases and controls, so they are not poolable. The MESA heart-failure preprint reports biomarker HRs only (PM_{2.5} is a co-adjustment), and the Ouagadougou OR (1.70) has no CI; neither enters the panel.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Cardiovascular & metabolic

1 record

Rivera-Mariani et al. 2026 medRxiv (preprint)

Inflammatory Burden, Structural Exposure, and Racial Disparities in Incident Heart Failure: Evidence from the Multi-Ethnic Study of Atherosclerosis

6,785 MESA participants followed ~16 years (485 HF events). IL-6, D-dimer, CRP and fibrinogen each predicted HF (HR 1.13-1.20 per SD); a composite index gave HR 1.25 (1.12-1.39); biomarker adjustment reduced the Black-White log-hazard gap by 57.5%. Associations were robust to PM_{2.5}/NO₂ co-adjustment in a geocoded subcohort. Limitation for this watch: air pollution is a sensitivity covariate only - no PM effect estimate is reported, and 'structural exposure' is inferred from a null interaction, not measured; single baseline biomarkers; preprint. Relevance: low - admitted because it tests whether PM explains part of an inflammatory pathway and reports that it does not change the estimates.

doi: 10.64898/2026.09.28.26364238

Respiratory & allergic

1 record

Zouma et al. 2026 Toxicol Rep

Pulmonary function assessment of a community exposed to air pollution in an industrial area in Ouagadougou, Burkina Faso

146 industrial-area residents vs 146 controls (Nioko 2), Nov-Dec 2023, with ATS/ERS spirometry. Symptom prevalence was higher in the industrial area (lower airway 89% vs 64%); pre-bronchodilator FEV1 %pred and FET% were lower ($p = 0.002$); restrictive defects were more common (OR 1.70, $p = 0.035$, no CI given), especially in women. Limitation: exposure is residence only - no PM or gas measurement, unadjusted univariate OR, and occupation, biomass cooking and socioeconomic differences between districts are uncontrolled. Relevance: moderate - West African industrial districts have essentially no PM monitoring; this is the kind of study a few calibrated nodes would turn from a residence contrast into an exposure-response.

doi: 10.1016/j.toxrep.2026.102344

Reproductive & developmental

1 record

Kukshal et al. 2026 Indian J Pediatr

Maternal Exposure to Air Pollution During Early Pregnancy and Congenital Heart Disease: An Exploratory Case-Control Analysis from Delhi NCR

1,556 participants (1,115 CHD cases, 441 controls) with residential exposure at conception and one month either side; multinomial regression, Bonferroni-corrected. Post-conception PM₁₀ was associated with all CHD (AOR 3.36, 2.05-5.52) and VSD (AOR 4.35, 2.41-7.83); preconception PM₁₀ with simple CHD (2.82, 1.69-4.71); 'poor AQI' gave AORs of 2.5-4.0. Limitation: the PM₁₀ contrast (per unit or category) is not given in the abstract; cases outnumber controls 2.5:1, implying hospital controls; Delhi-NCR monitor density makes residence-level contrasts small relative to city-wide covariance; ORs of 3-4 for an environmental exposure are far above the literature and suggest selection or confounding by SES. Relevance: moderate - a case for dense sensing where between-residence contrasts are the identifying variation.

doi: 10.1007/s12098-026-06363-x

Mechanistic toxicology

2 records

Saveleva et al. 2026 Mol Neurobiol

Urban PM0.2 Affects Astrocytes by Inducing Xenobiotic and Oxidative Stress Responses

Primary astrocytes exposed to collected urban PM0.2 at 50 ug/ml for 24 h: 2,217 mRNAs altered, with strong induction of xenobiotic metabolism (CYP genes, Ahrr), Nrf2 oxidative stress responses and glutathione detoxification (GST superfamily, Nqo1); 354 proteins nominally changed, 142 concordant with mRNA. Reanalysis of published data shows microglia go pro-inflammatory while astrocytes go detoxification/antioxidant. Limitation: one dose, one time point, 50 ug/ml is orders above any plausible brain dose, no particle-free extract control for organics vs core; proteomics is labelled exploratory. Relevance: low-moderate - reinforces that ultrafine/quasi-ultrafine number, not PM_{2.5} mass, is the metric for neuro endpoints, which most LCS do not resolve.

doi: 10.1007/s12035-026-06232-w

Ferreira Vicente et al. 2026 Expert Rev Respir Med

Pulmonary black carbon accumulation: mechanisms, health impacts, and environmental sources

Narrative review (searches to June 2026) of BC sources, physicochemistry, exposure assessment, biodistribution and tissue retention, with emphasis on tissue-based biomonitoring and label-free optical detection, and of respiratory, cardiovascular, neurological, developmental, metabolic and immune associations. The argument: tissue BC load could bridge external exposure estimates and biologically effective dose. Limitation: narrative, no quantitative synthesis or risk-of-bias assessment; tissue BC integrates an unknown exposure history, so it is a cumulative dose marker, not a validation of any particular exposure model. Relevance: low-moderate - internal BC dose is a possible ground truth for personal aethalometer and network-derived BC exposure.

doi: 10.1080/17476348.2026.2740205

Exposure assessment & modelling

5 records

Sheidaei et al. 2026 Environ Sci Technol*A Multi-Stage Deep Learning Framework for Daily PM_{2.5} Estimation at 100 m Resolution Across the Contiguous United States, 2000-2024*

Three-stage pipeline: a reanalysis-informed U-Net fills missing satellite AOD, a bidirectional LSTM refines its temporal structure and terrain downscales it to 100 m; a multistream network then predicts daily PM_{2.5} over >766 million cells for 2000-2024. Site-held-out validation: R² = 0.82, RMSE = 2.85 $\mu\text{g m}^{-3}$, MAE = 1.84 $\mu\text{g m}^{-3}$; spatial R² 0.94, temporal R² 0.78. Limitation: the 100 m detail comes from downscaled AOD and land-use covariates, not from 100 m observations - validation sites are regulatory monitors, which cannot test sub-kilometre gradients near roads or smoke plumes; the temporal R² (0.78) is the binding figure for short-term epidemiology, and performance during wildfire days is not reported. Relevance: high - a 25-year 100 m baseline is the natural prior that dense low-cost networks should be fused against, and the place their residuals are informative.

doi: 10.1021/acs.est.6c05375

Mielnik et al. 2026 Research Square (preprint)*Comparing geospatial interpolation techniques for estimating ambient PM_{2.5} mass concentrations from sugarcane burning in South Florida*

Low-cost sensors at 46 locations (8 Jun 2024 - 1 Dec 2025) spanning burn and non-burn seasons in the Everglades Agricultural Area; IDW and kriging variants compared by leave-one-out cross-validation for 24 h PM_{2.5}. Daily R² ranged 0-0.99; IDW won on most days, and a per-day best-model ensemble was most robust. Smoke presence raised estimated PM_{2.5} by 1.65 $\mu\text{g m}^{-3}$ (SE 0.13), concentrated inland near burn areas; the coast was cleanest. Limitation: sensor type, calibration and humidity correction are not stated in the abstract - in smoke, optical sensor bias is composition-dependent, so interpolation error and sensor error are confounded; LOOCV on a network this sparse rewards smoothness; preprint. Relevance: high - a direct test of how far a rural LCS network can be stretched into a health-study exposure surface for an episodic agricultural source.

doi: 10.21203/rs.3.rs-10743669/v1

Akhtar et al. 2026 Environ Monit Assess*Spatiotemporal characteristics of ozone and PM_{2.5} pollution in Jilin City: insights from observations in 2020 and 2023*

WRF-CMAQ at 3 km plus surface observations compare May-July 2020 (COVID low-emission baseline) with 2023 in a petrochemical city in Northeast China. Meteorology explains ~35% of O₃ variability (temperature-dominated) but <1% of PM_{2.5}; PM_{2.5} rose in May and fell sharply in July, staying high in Huadian, Chuanying and Jiaohe. Limitation, stated by the authors: emissions and meteorology both changed between the two years, so attribution is associational, and O₃ is systematically underestimated; <1% explained PM_{2.5} variance means the chosen predictors simply do not describe PM_{2.5} here. Relevance: low - mid-size industrial cities are under-monitored, but this adds two seasons, not a method.

doi: 10.1007/s10661-026-15929-3

Vukic et al. 2026 Environ Monit Assess*Integrated assessment of road and maritime traffic impacts on urban air quality and noise in a port-city environment*

Ten sites in Split, Apr-Sep 2024: 154 port and 66 road short-duration observations of PM_{2.5}, PM₁₀, CO, traffic counts and (road only) noise. Port means exceeded road means (PM_{2.5} 14.77 vs 11.64; PM₁₀ 18.25 vs 14.70 $\mu\text{g m}^{-3}$) but not significantly ($p = 0.22, 0.23$). Vessel count was weakly associated with PM₁₀ (R² = 0.082) and stayed positively associated with port PM_{2.5}/PM₁₀ in cluster-robust models; wind speed lowered PM. Limitation: spot measurements with uncalibrated low-cost sensors, no reference co-location, and vessel count ignores engine type, berth time and fuel - the R² says ship numbers are a poor proxy for ship emissions. Relevance: moderate - the authors frame LCS correctly as screening-level; port sources need continuous nodes with plume-resolving time resolution, not spot visits.

doi: 10.1007/s10661-026-15986-8

Nassir et al. 2026 Environ Pollut*A multi-sensor satellite assessment of air quality during the 2026 Middle East Conflict across nine regional countries*

A dual-baseline design (14-month annual and seasonally matched prior year) applied to the 39-day 2026 conflict period across nine countries (~217 million people). Domain-mean anomalies: NO₂ -15.1%, SO₂ -45.6%, CO -8.6%, AOD -19.6%, CH₄ +0.37% (noise), ordered by atmospheric lifetime - the signature of an emission change; cold spots sit over Iranian, Kuwaiti and Qatari hydrocarbon fields. The NO₂ burden estimate (-7,849; 95% CI -35,641 to +13,551) crosses zero, and the PM_{2.5} burden is called indeterminate because dust and anthropogenic aerosol cannot be separated at CAMS resolution. Limitation: 39 days, no surface PM validation; a wind reversal is documented. Relevance: moderate - an honest statement that column AOD cannot apportion PM in dust-dominated regions without ground speciation.

doi: 10.1016/j.envpol.2026.129258

Sensing, forecasting & instrumentation

2 records

Yan et al. 2026 Atmosphere*Baseline Determination for Threshold-Based Interpretation and Groups of Single-Particle Aerosol Fluorescence Measurements*

Tests deionised water, NaCl and SiO₂ as low-fluorescence baselines for the SwisensPoleno Jupiter and compares $\mu + 3\sigma$, P99 and MAD thresholds. NaCl gives the broadest baseline ($\mu + 3\sigma$ upper thresholds ~52.0, 39.3, 38.2 $\times 10^{-3}$ a.u. across excitation sources, vs 6.6/10.1/3.1 for H₂O and 12.1/9.1/2.6 for SiO₂), and the effect persists within a shared 10-30 μm size range; baseline choice changes the fraction of fluorescence-positive particles in both test materials and summer/winter ambient data. P99 tracks $\mu + 3\sigma$; MAD diverges. Limitation: one instrument model, and no ground truth for which threshold best recovers biological particles. Relevance: moderate - fluorescence-positive counts from different networks are not comparable unless baseline material and threshold rule are reported, which is a metadata standard LCS bioaerosol channels lack.

doi: 10.3390/atmos17100971

Yang et al. 2026 Environ Sci Technol Lett*Development and Diagnosis of a Temporal-Context U-Net Deep Learning Surrogate Model for CMAQ Particulate Nitrate over China*

An attention U-Net takes the target hour, the previous 2 h and a 96 h summary to emulate CMAQ surface nitrate over China; it beats baseline surrogates, especially at high concentrations. Emission-reduction tests reproduce the direction of July national-mean nitrate decreases but underestimate their magnitude, and agreement is poor for January NO_x responses outside training months. Residuals correlate with CMAQ NH₃ and particle H⁺, and adding them helps. Limitation: it emulates a model, not the atmosphere - CMAQ's own nitrate biases are inherited - and the weak emission-response skill is exactly what a policy surrogate needs. Relevance: moderate - nitrate is the PM component optical sensors misread most under humidity, and fast surrogates are a route to composition-aware calibration.

doi: 10.1021/acs.estlett.6c00755

Occupational & indoor

2 records

Charres et al. 2026 Atmos Pollut Res*Insights into the organic and toxicological characteristics of PM₁₀ from indoor and outdoor settings in a school environment*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 2 Oct with no abstract; Europe PMC and Semantic Scholar returned nothing and Scholar Gateway is still disconnected. The title describes organic and toxicological characterisation of indoor and outdoor PM₁₀ at a school; the author list (Cipoli, Vicente, Evtugina) is the Aveiro aerosol group, but this watch does not infer location or method from authorship. Relevance: potentially high - indoor/outdoor school PM composition is what classroom low-cost sensors cannot report. On the retry list.

doi: 10.1016/j.apr.2026.103223

Parthiban et al. 2026 Cureus*Ocular Surface Disorders in Traffic Police Officers in the Union Territory of Puducherry, India: A Cross-Sectional Study*

106 traffic police examined with OSDI, Schirmer, TBUT, tear meniscus height and conjunctival impression cytology: Nelson grade 1-3 metaplasia in 83%, dry-eye symptoms in 78%; grade associated with years of field duty, OSDI and TBUT (chi-square $p < 0.05$). Limitation: no unexposed comparison group, no exposure measurement, UV and pollution are inseparable, and years of service is collinear with age; the authors' 'clear exposure-outcome relationship' is not supported by this design. Relevance: low - a case for wearable PM monitoring in traffic-police cohorts, which would make this question answerable.

doi: 10.7759/cureus.115533

Burden, policy & mitigation

1 record

Yang et al. 2026b Environ Res*Process-resolved characterization and plant-level inventory of multi-pollutant emissions from China's coking industry*

Stack measurements at coal charging, coke-oven flue gas, dry quenching and pushing in two plants (total PM 3.12-37.72 mg/m³; sulfate 28-59% of ions; As and Cd enriched in PM_{2.5}; the condensable fraction carries 44.5-57.4% of trace elements in flue gas) feed a 307-plant inventory for 2012-2024. 2022 national emissions: 159.8 kt PM, 66.6 kt SO₂, 82.3 kt NO_x, 838.3 kt VOCs, 4.24 kt PAHs, 707.6 t trace elements; BACT cuts PM to 15.5 kt by 2050. Limitation: two plants anchor factors for 307; fugitive emissions, flagged as a priority, are the least measured; condensable PM is operationally defined. Relevance: moderate - condensables are invisible to stack filters but appear as ambient PM_{2.5} downwind, where fence-line sensor networks see them.

doi: 10.1016/j.envres.2026.125824

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Rivera-Mariani et al.	Inflammatory Burden, Structural Exposure, and Racial Disparities in Incident Heart Failure: Evidence from the Multi-Ethnic Study of Atherosclerosis	<i>medRxiv (preprint)</i>	Prospective cohort	PM _{2.5} and NO ₂ as co-adjustment in a geocoded subcohort (not the exposure of interest)	USA (MESA)
Reproductive & developmental					
Kukshal et al.	Maternal Exposure to Air Pollution During Early Pregnancy and Congenital Heart Disease: An Exploratory Case-Control Analysis from Delhi NCR	<i>Indian J Pediatr</i>	Case-control	Periconceptional PM _{2.5} , PM ₁₀ , NO ₂ , CO, SO ₂ , O ₃ , AQI at geocoded residence	India (Delhi NCR)
Respiratory & allergic					
Zouma et al.	Pulmonary function assessment of a community exposed to air pollution in an industrial area in Ouagadougou, Burkina Faso	<i>Toxicol Rep</i>	Observational - cross-sectional	Residence in an industrial area vs control district (no PM measured in abstract)	Burkina Faso (Ouagadougou)
Mechanistic toxicology					
Ferreira Vicente et al.	Pulmonary black carbon accumulation: mechanisms, health impacts, and environmental sources	<i>Expert Rev Respir Med</i>	Narrative review	Black carbon: sources, tissue retention, label-free optical detection	Not stated (review)
Saveleva et al.	Urban PM _{0.2} Affects Astrocytes by Inducing Xenobiotic and Oxidative Stress Responses	<i>Mol Neurobiol</i>	In vitro exposure	Urban ambient PM _{0.2} (aerodynamic diameter <0.2 µm), 50 µg/ml, 24 h	Not stated (laboratory)
Occupational & indoor					
Charres et al.	Insights into the organic and toxicological characteristics of PM ₁₀ from indoor and outdoor settings in a school environment	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - indoor/outdoor school PM ₁₀ organic speciation and toxicity (by title)	Not recoverable from metadata
Parthiban et al.	Ocular Surface Disorders in Traffic Police Officers in the Union Territory of Puducherry, India: A Cross-Sectional Study	<i>Cureus</i>	Observational - cross-sectional	Occupational traffic exposure (>1 year field duty); no PM measured	India (Puducherry)
Sensing, forecasting & instrumentation					
Yan et al.	Baseline Determination for Threshold-Based Interpretation and Groups of Single-Particle Aerosol Fluorescence Measurements	<i>Atmosphere</i>	Laboratory sampler validation	Single-particle fluorescence (SwisensPoleno Jupiter), bioaerosol 10-30 µm	Ireland (laboratory + ambient case studies)
Yang et al.	Development and Diagnosis of a Temporal-Context U-Net Deep Learning Surrogate Model for CMAQ Particulate Nitrate over China	<i>Environ Sci Technol Lett</i>	Machine-learning model	Hourly surface particulate nitrate, CMAQ emulation (July/December 2011-2020)	China (national)
Exposure assessment & modelling					
Akhtar et al.	Spatiotemporal characteristics of ozone and PM _{2.5} pollution in Jilin City: insights from observations in 2020 and 2023	<i>Environ Monit Assess</i>	Chemical transport model	Warm-season PM _{2.5} and O ₃ , 3 km WRF-CMAQ + surface monitors, meteorological normalised	China (Jilin City)
Mielnik et al.	Comparing geospatial interpolation techniques for estimating ambient PM _{2.5} mass concentrations from sugarcane burning in South Florida	<i>Research Square (preprint)</i>	Statistical model	Daily PM _{2.5} from a dense low-cost sensor network (46 sites), IDW vs kriging	USA (Florida, Everglades Agricultural Area)

First author	Focus	Journal	Design	Metric	Region
Nassir et al.	A multi-sensor satellite assessment of air quality during the 2026 Middle East Conflict across nine regional countries	<i>Environ Pollut</i>	Satellite time-series analysis	AOD (MODIS MAIAC), TROPOMI/OMI NO ₂ , SO ₂ , CO, CH ₄ ; CAMS PM _{2.5} (indeterminate)	Middle East (nine countries)
Sheidaei et al.	A Multi-Stage Deep Learning Framework for Daily PM _{2.5} Estimation at 100 m Resolution Across the Contiguous United States, 2000-2024	<i>Environ Sci Technol</i>	Machine-learning model	Daily PM _{2.5} surface, 100 m, from gap-filled MAIAC-type AOD + meteorology + geospatial predictors	USA (contiguous)
Vukic et al.	Integrated assessment of road and maritime traffic impacts on urban air quality and noise in a port-city environment	<i>Environ Monit Assess</i>	Field measurement	Short-duration low-cost PM _{2.5} /PM ₁₀ and CO at 10 sites; vessel and vehicle counts	Croatia (Split)
Burden, policy & mitigation					
Yang et al.	Process-resolved characterization and plant-level inventory of multi-pollutant emissions from China's coking industry	<i>Environ Res</i>	Modelling / inventory	Size-segregated stack PM, ions, trace elements; plant-level inventory 2012-2024, projections to 2050	China (national, 307 plants)

Provenance and method

Entry window **2 October 2026**, single day, continuous with the 1 October issue. Harvester legs run leg-by-leg: **PubMed** health 6, sensing 1; **Europe PMC** 31; **Crossref** by ISSN 40; **arXiv** no entry dated 2 October; **OpenAlex** HTTP 429. 78 raw deduplicated to **76, 9** already carried, **67** fresh. The **PubMed connector** ([EDAT] 2 October) returned 7 health and 0 sensing PMIDs; 4 appended. **71** candidates screened, **15 admitted**, **56 rejected** with logged reasons (*AMT/ACP* cloud, ozone and radiation work, *MST* engineering, *ES&T* water and soil chemistry, LCA records, a benzene-only cohort, NO₂-only trends, an editorial and a commentary). **Consensus** returned 10 calibration hits, **0 new** (all archived or previously rejected). **ClinicalTrials.gov**: no PM/air-pollution registration updated 2–3 October. **Scholar Gateway** still fails with an identity error. **One record carried on metadata alone**. DOI gate: **0 fail, 0 warn**. Abstracts remain the copyright of their respective publishers.